

Introduction / Purpose

This project was created to support my job search after graduation. I focused on the two areas where I am most likely to relocate: Bethesda/DC and Philadelphia, PA. To compare the entry-level technology job markets in these areas, I collected job listings from LinkedIn, Indeed, and several company career sites.

Data Collection and Methodology

For each listing, I reviewed the job description and recorded important details such as role type, required skills, salary range, experience requirements, degree requirements, work arrangement, security clearance requirements, and U.S. citizenship requirements. I then organized the data into a main spreadsheet and used PivotTables and charts to identify patterns in job availability, role categories, salary information, technical skill demand, and clearance or citizenship barriers.

Dataset Overview

The dataset includes 65 main job listings, with 35 from the Bethesda/DC area and 30 from the Philadelphia area. I also included a separate sheet for stretch roles, which are positions that were somewhat outside my main entry-level criteria, required more experience, required active security clearances, or were less realistic from a commuting standpoint. Keeping these roles separate allowed me to preserve them for reference without distorting the main entry-level comparison.

The first chart compares the total number of job listings collected from each metro area. For the purpose of my own job search, this provides a useful comparison between the two areas I am most likely to move to after graduation. At a larger scale, this same approach could also be used to compare additional metro areas and help Computer Science graduates evaluate where they may want to pursue entry-level technology roles.

Role Category Comparison

The role comparison showed that Bethesda/DC had slightly more listings overall, but the role categories were not evenly distributed between the two markets. Bethesda/DC had more IT support, infrastructure, government contractor, and clearance-related roles, while Philadelphia had stronger representation in software, application support, healthcare IT, fintech, and corporate analytics roles.

While collecting listings manually and attempting to balance the dataset across role categories, I noticed that many additional postings began to fit better in the stretch role category rather than the main entry-level dataset. These roles often required active security clearances, more experience, or a less realistic commute. Because the data was collected by hand, the sample size is limited and may not fully represent the entire job market in either area. With a

larger sample size or an automated job collection program, the role distribution could be analyzed more accurately and at a greater scale.

Salary Comparison

I created an average salary chart and table to compare roles that included listed salary ranges. Salary information was available for 25 Bethesda/DC listings and 13 Philadelphia listings, so the salary comparison is based only on those postings rather than the full dataset. For each posting with a salary range, I calculated the midpoint by averaging the minimum and maximum salary values. I then used those midpoint values to compute the average listed salary for each metro area.

This comparison should be treated as a rough estimate rather than a complete salary analysis because not every job posting included salary information. Based on the listings that did include salary ranges, Bethesda/DC had an average listed salary of \$70,783.92, while Philadelphia had an average listed salary of \$70,614.77. The difference between the two averages was very small, suggesting that the two areas were fairly similar in listed compensation within this sample. Because the salary averages were so close, role type, commute, clearance requirements, and accessibility may be more important factors than salary alone when comparing these two markets.

Skill Demand Comparison

The skill demand chart compares the technical skills and tools that appeared most often across the job listings. SQL, Python, Excel, ticketing systems, cloud tools, and Power BI were some of the most common skills across the dataset. Data and analytics roles tended to emphasize SQL, Python, Excel, Power BI, reporting, dashboards, and data visualization. IT support roles more often emphasized ticketing systems, troubleshooting, Microsoft 365, Active Directory, and cloud-based user support tools.

The comparison also showed differences between the two metro areas. Bethesda/DC had more listings connected to clearance-eligible government work, cloud infrastructure, technical support, and federal contractor environments. Philadelphia showed strong demand for data analysis, software and application support, healthcare IT, fintech, and business reporting tools. This suggests that while both areas value technical skills, Bethesda/DC may be more infrastructure- and government-focused, while Philadelphia may offer a broader mix of corporate, healthcare, finance, and application-focused technology roles.

Clearance and Citizenship Barriers

One major difference between the two markets was the presence of clearance and citizenship barriers. In the main dataset, 11 of the 35 Bethesda/DC listings had a clearance or citizenship barrier, compared with 3 of the 30 Philadelphia listings. Since the Bethesda/DC area has more federal contractors, defense organizations, intelligence-related employers, and

government customers, more job listings in that area mentioned U.S. citizenship requirements, clearance eligibility, or active security clearance requirements. Philadelphia had fewer of these barriers, with more listings connected to healthcare, finance, education, local government, corporate IT, and software/application support.

For my personal job search, the U.S. citizenship requirement is not a major barrier. However, active security clearance requirements could still limit opportunities, as some roles require candidates to already hold a clearance before applying. This means Bethesda/DC may offer more government-adjacent technology opportunities, but some of those roles may be harder to access without an existing clearance. Philadelphia may offer fewer total listings in this sample, but the roles appear more accessible overall for entry-level candidates without clearance.

Personal Career Fit

Based on my background in Computer Science, this dataset helps me identify which roles are the best fit for my education, skills, and career goals. The job listings gave me a clearer idea of what employers are looking for in entry-level technology candidates, including common requirements such as SQL, Python, Excel, Power BI, ticketing systems, cloud tools, troubleshooting, and communication skills.

This project also helps me better understand what to expect from the job market in both Bethesda/DC and Philadelphia. Since I may end up living in both areas at different points, comparing these markets gives me a practical way to plan ahead. Bethesda/DC appears to offer more government-adjacent roles, especially in IT support, infrastructure, and data/reporting. Philadelphia appears to offer a strong mix of data analytics, software/application support, healthcare IT, fintech, and corporate technology roles. Overall, the dataset helps me narrow down which types of roles to target and which skills to continue building before graduation.

Conclusion and Future Improvements

Overall, this project showed that both Philadelphia and Bethesda/DC have realistic entry-level opportunities for graduating Computer Science majors. The comparisons helped identify which types of roles are more common in each area, what skills appear most often in job descriptions, and which requirements may affect accessibility, such as clearance or citizenship barriers.

This project was useful for my own job search because it helped clarify which roles best match my background and which skills I should continue developing. It also showed that salary alone may not be the deciding factor, since the average listed salaries in both areas were very similar. Instead, role type, required skills, commute, and accessibility are important factors to consider when choosing where to pursue a career.

With more time, data, and resources, this project could be expanded into a larger personal job-finding tool. A future version could collect listings automatically, compare more cities, filter by major or career path, and help Computer Science students, as well as students in other fields, better understand where to begin their careers after graduation.